Tourism Management 71 (2019) 179–186

Contents lists available at ScienceDirect

Tourism Management

journal homepage: www.elsevier.com/locate/tourman

Recreation and environmental quality of tropical wetlands: A social media based spatial analysis☆

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ARTICLE INFO

Keywords:
Recreational ecosystem services
Geotagged photographs
Passive crowdsourcing
Social media
Tourism
Water quality
Wetland ecosystems

ABSTRACT

Here we test an application, based on the analysis of the metadata of geotagged photographs, to investigate the provision of recreational services by the network of wetland ecosystems in the state of Kerala, India. We estimate visitation to individual wetlands state-wide and extend, for the first time to a developing region, the emerging application of recreational ecosystem services (ES) modelling using data from social media and environmental quality data. The impacts of restoration of wetland areal extension and water quality improvement are explored as a means to inform sustainable management strategies. Findings show that improving water quality to a level suitable for the preservation of wildlife and fisheries could increase annual recreational visits by 13% (350,000) to wetlands state-wide. Results support the notion that passive crowdsourced data from social media has the potential to improve current ecosystem service analyses and environmental management practices also in the context of developing countries.

1. Introduction

Crowdsourced data represent an opportunistic source of behavioural data for ecosystem services (ES) analysis. The use of such data allows for innovations that extend the scale and improve the timespan within which ES are assessed. This is especially relevant for developing regions, where ES analysis is often limited by data and resource scarcity. This research uses the tropical wetlands of Kerala state in India as a case study through which to explore one such innovation in the form of a spatial analysis of recreation using geotagged data from social media and environmental parameters.

Wetland ecosystems support a rich biological diversity and provide valuable cultural services to human beings by offering spaces for recreation, promoting mental and physical health, catalysing tourism or inspiring culture, spiritual experiences and sense of place (de Groot et al., 2012). Their benefits are however often not captured by market mechanisms leading their services to go undervalued (or unvalued) (Ghermandi, van den Bergh, Brander, de Groot, & Nunes, 2010). Without adequate understanding and characterization of these ES, there is a high risk that these unique ecosystems will not receive the appropriate level of protection and conservation to ensure their future ability to support them. Although increasing numbers of studies focus on wetland ecosystems and their benefits (Brander, Florax, & Vermaat, 2006; Brouwer, Langford, Bateman, & Turner, 1999; Ghermandi et al., 2010; Woodward & Wui, 2001), ES valuations of wetlands in developing regions are inadequately represented (Chaikumbung, Doucouliagos, & Scarborough, 2016). Valuation of cultural ES in particular are challenging (Satz et al., 2013) and make up only a small fraction of the total studies. Particularly in India, only a limited number of studies focussing on wetland ES are available to date (Ghermandi, Sheela, & Justus, 2016) in spite of its diverse wetland habitats and the substantial stress they experience from anthropogenic sources (Banerjee & Dey, 2017; Bassi, Kumar, Sharma, & Pardha-Saradhi, 2014; Parikh & Datye, 2003).

A common way to investigate the impact of changes in wetland quality on the provision of cultural ES such as recreation and ecotourism relies on surveys, which can be performed either on-site or off-site to explore individuals' or households' stated or revealed preferences under various environmental quality scenarios (Bhatta et al., 2016). These methods are however costly, time consuming, and often need to rely on the visitors' recollection of past behaviour to analyse temporal trends. An emerging approach, which may overcome some of the aforementioned limitations, uses spatial data from social media to investigate people's attitudes and behaviour with regards to natural

☆ This research (Grant No. 2751/16) was supported by the Israel Science Foundation within the ISF-UGC joint research program framework.
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https://doi.org/10.1016/j.tourman.2018.10.018
Received 1 August 2018; Received in revised form 11 October 2018; Accepted 12 October 2018
Available online 17 October 2018
0261-5177/

ecosystems. In particular, metadata contained within geotagged photos from photo-sharing websites have been used to understand interactions between humans and the natural environment at wide spatial scales and with low resource requirements (Wood, Guerry, Silver, & Lacayo, 2013).

The global penetration of smart phones and the integration of Global Positioning System (GPS) technology have led to vast amounts of photographs that are geotagged with location information. Moreover, the development of social media networks, including photo-sharing websites such as Flickr (http://www.flickr.com), makes publicly available a wealth of user-provided, geotagged metadata for retrieval and analysis. Geotagged photo counts have been proposed as a proxy source of revealed information regarding patterns of public use and visitation frequency, for nature-based recreation in particular (Chua, Servillo, Marcheggiani, & Moere, 2016; Figueroa-Alfaro & Tang, 2017; Hausmann et al., 2017; Mancini et al., 2016; Sinclair, Ghermandi, & Sheela, 2018; Tenkanen et al., 2017; Wood et al., 2013) Popular photo-sharing websites (such as Flickr, Panoramio and Instagram) have been used as a source for information on recreational visits in wetland and freshwater ecosystems (Ghermandi, 2016; Keeler et al., 2015), marine and coastal ecosystems (Howarth, 2014; Sinclair et al., 2018), protected areas (Walden-Schreiner, Rossi, Barros, Pickering, & Leung, 2018) and urban areas (Sun, Fan, Helbich, & Zipf, 2013; Vu, Li, Law, & Ye, 2015). Such research shows that spatiotemporal geotagged data from social media can be successfully used as a proxy for recreational visits, to estimate recreational value (Sinclair et al., 2018) and, in combination with historic environmental quality data, may allow for the exploration of how visitation would be affected under different water quality scenarios (Keeler et al., 2015). Being available retrospectively and at low cost, such a technique may be particularly useful to investigate nature-based recreation patterns and ecosystem quality at wide spatial scales in the context of developing countries, especially in contexts where, like in the state of Kerala, the tourism industry is growing rapidly but the local natural capital is undergoing substantial depletion.

The present paper describes the assessment and mapping of the recreational benefits of wetlands in Kerala based on the analysis of geotagged photos from the popular photo-sharing website Flickr. The study first uses collected visitation data from 42 areas in Kerala to test the performance of a proxy model for geotagged photo counts as a surrogate for visitation. The results of this analysis are used to infer visitation rates for all the major wetlands in Kerala. Second, the relationship between estimated wetland visitation rates and various wetland attributes, including water quality and area, are investigated for the 81 wetlands in Kerala for which water quality data could be retrieved. Third, prospective changes in visitation frequencies are explored relative to improvements in water quality for 48 wetlands with currently poor water quality and changes in wetland area for 24 coastal lagoons and inland open waters. The study concludes by discussing the implications of the results in terms of promoting a more sustainable management of Kerala's water ecosystems and the potential of the proposed technique for application in other geographic and socio-economic contexts.

2. Materials and methods

2.1. Study area

Kerala state is located in the southernmost part of India and embraces the coast of the Arabian Sea on the west and the Western Ghats on the east (Fig. 1). Kerala has the largest proportion of land area under wetlands among all the states of India (Abraham, 2015). Its lakes and backwaters are internationally renowned for their uniqueness and aesthetic value and make Kerala one of India's top tourist destinations. Kerala's wetland ecosystems are estimates to cover 160,590 ha and include three Ramsar sites. Major wetland types in Kerala are rivers (65,162 ha), lagoons (38,442 ha), reservoirs (26,167 ha), waterlogged areas (20,305 ha), as well as beaches, canals and ponds (Government of India, 2010). Fig. 1 shows the spatial distribution of Kerala's wetlands included in the National Wetland Atlas of Kerala (Government of India, 2010).

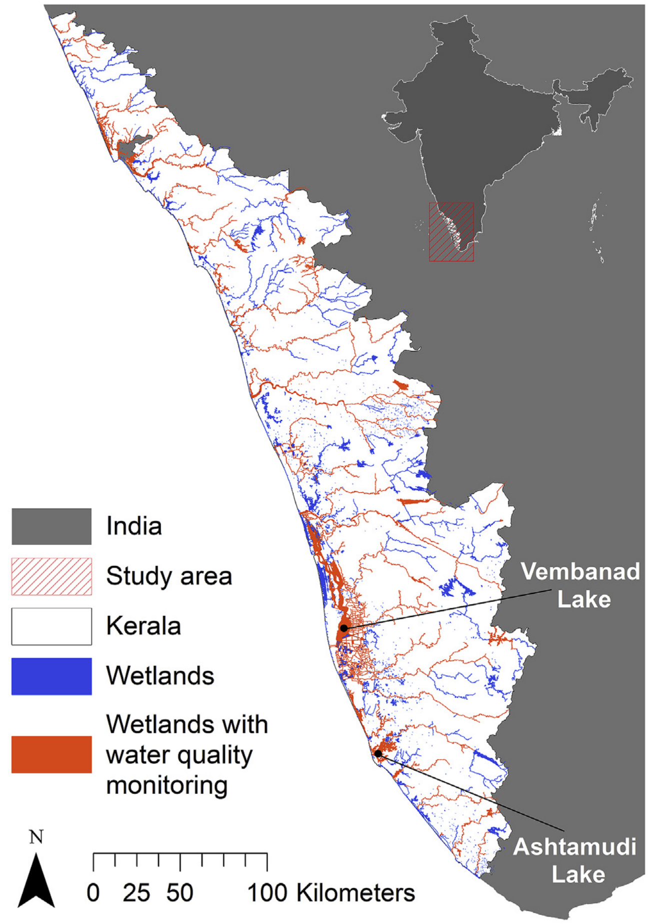

Fig. 1. Location of Kerala state and its wetlands.

Kerala is renowned for its backwaters, perennial or temporary estuaries making up around 42,000 ha across 30 sites (Thomson, 2002). These unique wetland ecosystems have historically played a vital role in sustaining local communities and more recently play a central role in a growing tourism industry. Foreign and domestic tourism in Kerala has constantly increased between 2005 and 2016, from just over 6 million in 2005 to over 14 million in 2016, with tourism earnings amounting to 296.5 billion rupees ($4.5 billion) in 2016 alone (Kerala Department of Tourism, 2016). The state's unique wetland ecosystems are recognized as a major draw for tourists, who frequent the wetlands primarily for boating purposes and appreciation of the aesthetic value of the unique landscape and the local culture and biodiversity (Vincy, Rajan, & Kumar, 2012).

Kerala's wetlands are however subjected to substantial anthropogenic pressures from fast growing urbanization, tourism development, agricultural intensification, and land reclamation (Shiji, Sabitha, Prabhakar, & Harikumar, 2016). This can be partly attributed to the inadequacy in accounting for the benefits of the environmental resources they provide to the local communities (Varkey, Kumar, Mridha, Sekhar, & Sahoo, 2016) and may compromise their future ability to sustain healthy ecosystems and the provision of ES. Such deterioration comes in the form of water quality reduction from pollution, eutrophication and mining activities, but also as encroachment and reclamation of lake area (Kokkal, Harinarayanan, & Sahu, 2007; WISA, 2013). The Kerala State Pollution Control Board (KSPCB), which is responsible for monitoring water quality in Kerala's wetlands, observes water quality deterioration in the majority of wetlands between 2008 and 2016 (Kerala State Pollution Control Board, 2016). The 81 wetlands for which water quality monitoring is available from KSPCB are highlighted in orange in Fig. 1. Encroachment and reclamation of lakes is driven by aquaculture, coir industries, expansion of agriculture, as well as urban expansion and harbour development (Sitaram, 2014). Though comprehensive estimates of reclamation are not available at the state level, figures are available for two of the three wetlands in Kerala that are recognized under the Ramsar Convention on Wetlands of International Importance: Vembanad Lake, the largest wetland ecosystem in Kerala, has suffered reclamation of 6.92% between 1973 and 2015 (Nair and Babu, 2015) with an estimated 2% of its area lost in 2014 alone (Roopa & Vijayan, 2017); Ashtamudi Lake lost over 10% of its area between 1999 and 2006 (Sitaram, 2014). The problem of lake reclamation is exacerbated by its potential impact on water quality (Sitaram, 2014). Research has suggested that Kerala's wetland ecosystems are subjected to exploitation beyond their supportive capacity (Paul, Alias, & Varghese, 2014; WISA, 2013; WISA & CWRDM, 2017) and that the facilities necessary for tourism such as hotels, resorts and houseboats are being developed to meet demand without due concern for the natural system within which they reside. Tourism in the Vembanad ecosystem, for instance, is a source of pollution especially in the dry season when tourism demand is high and the system's assimilative capacity is low (Paul et al., 2014). Such inputs include treated and untreated sewage, deposits into the lake from surrounding settlements and houseboats, and pollution from the tourism port (Mathew & Swain, 2017; Safoora Beevi & Devadas, 2014). The future deterioration of water quality and loss of wetland extent may have a major impact on the aesthetic value of Kerala's wetlands and thus on recreational visitation, which is crucial for the revenue generated from the tourism industry on which the economy is becoming ever more dependent (Abraham, 2015; Thimm, 2017).

2.2. Geotagged photo counts as a proxy for visitation

As one of the largest sources of geotagged photographs available online, with an easy-to-access application programming interface (API), Flickr was used to recover 47,246 photographs from within Kerala's border. Using R scripts (R Develoopment Core Team, 2014) to interact with the Flickr API, metadata was extracted from photographs taken within a bounding envelope of Kerala state, between 1 January 2005 and 31 December 2016. Public metadata received included user id, photograph coordinates, and a time stamp from when the picture was taken. Geotagged photos were subsequently combined with spatial boundaries of the network of wetland sites in Kerala state as characterized by a multiple polygon shapefile, available from Government of India's National Wetland Atlas project (Government of India, 2010). Photographs taken at wetland sites were extracted if located within a 60 m buffer measured outwards from the wetland boundaries in order to account for photographs taken on the shores or banks of wetlands.

Photo-user-days (PUD), i.e., unique combination of user and recreation site (Tenkanen et al., 2017; Wood et al., 2013), were extracted from the dataset of photographs' metadata for each wetland to account for the fact that a single user can take multiple photographs during a single visit. The location of the first photograph taken by a user on a given day at a wetland was taken as representative for the PUD. PUD by a user at the same wetland on consecutive days were assumed as separate. Using this strategy, a PUD count was created for each wetland in the study area between 2005 and 2016.

In order to generate estimates of annual visits it was necessary to calibrate PUD counts as a proxy for visitation in the context of the study region (Sessions, Wood, Rabotyagov, & Fisher, 2016; Tenkanen et al., 2017; Wood et al., 2013). We first retrieved foreign and domestic overnight recreational visitation data from Kerala's department of tourism for 42 scenic (n = 33), cultural (n = 6) and urban (n = 3) sites throughout Kerala for 2014 (Kerala Department of Tourism, 2014). Scenic sites include coastal and mountain areas with nature reserves, hill stations, beaches, lakes, rivers and waterfalls. Second, we estimated day trips for each of these sites using the ratio to overnight visitors in 2014, the former having been obtained from the Government of India's 72nd National Sample Survey which surveyed 5843 households in Kerala between 2014/15 about their visitation behaviour (India Department of Economics and Statistics, 2015). A univariate, OLS model was administered for the sample of 42 sites (Ghermandi, 2016; Levin, Lechner, & Brown, 2017):

$$\log y_i = \beta_0 + \beta_1 \log x_i + \varepsilon$$

where y_i is the visitation rate for the ith site in 2014; x_i is the total PUD count for the ith site, β_0 and β_1 are the intercept and slope coefficient respectively; and ε is the error term. The assumption of normality for the distribution of residuals was tested with the Shapiro-Wilk diagnostic test. Following Ghermandi (2016), in predicting visitation rates from the log-transformed estimates, we consider both the naïve forecasts obtained by simply applying the exponential transformation to the logarithmic estimates, the theoretically optimal forecasts under error normality assumptions (Bårdsen & Lütkepohl, 2011), and the estimates obtained with Duan's smearing technique, which relaxes the assumption of normal distribution of the errors (Duan, 1993). The root mean squared error (RSME) and the mean absolute error (MAE) verification statistics were calculated for the 42 calibration sites. Combining the estimated visitation data with the boundaries of the multi-polygon shapefile of Kerala's wetlands in ArcGIS 10.2.2, enabled a yearly estimate of recreational visits to be mapped for Kerala.

2.3. Regression of visitation against site- and context-specific variables

The relationship between PUD and wetland attributes was explored for 81 wetlands in Kerala, which were chosen based on the availability of water quality data (Fig. 1). A range of water quality measures for the years 2008–2016 for wetlands in Kerala are available in print from KSPCB (2016). KSPCB's water quality measures were classified ranging from 1 (= highest quality) to 6 (= lowest quality), which we regarded as a continuous variable. The limits classified by the KSPCB for swimming and for the preservation of fisheries and wildlife are 2 and 4 respectively. Water testing stations were mapped in GIS to attribute water quality to the relevant wetlands for the available years. In cases where a water body had a series of testing station in different locations the average distance between stations was used to partition the wetland polygon into subsections for analysis. This resulted in 81 wetlands split into 139 individual wetland sections which were suitable for analysis. In addition to water quality and compatibly with data availability, a range of additional variables were included, which were found in previous studies to affect recreational visits in wetland ecosystems (see, for instance, Ghermandi & Fichtman, 2015). These include specific characteristics of the site such as wetland area and wetland type as well as context-specific information to characterize the accessibility of a site to potential users and the size of the potential market of recreationists. The latter includes the mean elevation within 1 km of the wetland, the mean travel time to the nearest city with 50,000 residents (Nelson, 2008), the population within 5 km and the abundance of wetland ecosystems within a 5 km radius.

PUD counts for each wetland were modelled against wetland attributes using a Negative Binomial (NB) regression, which can account for overdispersion in the data (Cameron & Trivedi, 2013), with forward mixed variable selection where variables were retained with a p-value of less than 0.05. The NB regression model was set up as follows:

$$\log y_i = \beta_0 + \beta_{SITE} x_{SITEi} + \beta_{POP} x_{POPi} + \beta_{WQ1} x_{WQi} + \beta_{WQ2} x_{WQi}^2 + \beta_{DP} + \varepsilon$$

where y_i is the number of PUD for site i, β_0 represents the model

Table 1
Explanatory variables of the negative binomial regression model.

Group	Variable (unit)
Wetland attributes (x_{SITE})	Area (km^2)
	Elevation (m)
	Wetland type (coastal lagoons, inland open water, rivers)
	Wetland abundance within 5 km buffer (km^2)
Human attributes (x_{POP})	Travel time to nearest large city (minutes)
	Population within 5 km buffer
Water quality (x_{WQ})	Water quality (scale from 1 to 6)

intercept, β_{DP} represents the negative binomial dispersion coefficient, and ε is the error term. The explanatory variables were grouped into wetland and human attributes, as outlined in Table 1. The inclusion of both linear and quadratic terms for all variables (with the exception of the categorical variable "Wetland type") were explored in order to allow for non-linear relationships with visitation.

2.4. Assessing impacts of changes in wetland quality on recreation

To explore how potential changes in water quality and wetland area might affect recreational visitation for Kerala's wetlands, the coefficients for each variable from the regression equation were adjusted independently in order to assess the relative change in the dependent variable (number of PUD). Keeping all other variables constant (*ceteris paribus*), we considered for this analysis adjusting the water quality parameter of wetlands experiencing low or very low water quality (WQ = 5 and WQ = 6) to a level that KSPCB considers adequate for the preservation of fisheries and wildlife (WQ = 4) and assessed the subsequent change in visitation. Moreover, the sensitivity of visitation to changes in wetland area for all coastal lagoons and inland open waters was explored with specific focus on the two Ramsar wetlands, Vembanad Lake and Ashtamudi Lake, which have experienced major encroachment in recent years.

3. Results

3.1. Geotagged photographs as a proxy for visitation

Of the 47,246 Flickr photographs (corresponding to 5897 PUD) taken between 2005 and 2016 in Kerala, analysis in ArcGIS 10.2.2 revealed 13,784 photographs from the dataset within a 60 m buffer of the network of wetlands in the state. Further analysis revealed from these 13,784 photographs, 3770 PUD taken by 1066 unique visitors within 243 separate wetland sites (Fig. 2). This equates to a mean of 3.5 visits per user, each taking 3.6 photographs on average per visit. In order to assess the influence that buffer value has on PUD counts, a sensitivity analysis was performed. Reducing the buffer width to 30 m leads to a 25% decrease in the number of PUD; increasing it to 100 m leads to a 29% increase in the number of PUD.

In total, 2060 PUD from 8352 Flickr photographs were found within a 60 m buffer of the 139 wetlands for which water quality is available. Photographs were found within 81 of 139 wetland polygons. PUD were most densely concentrated in the districts of Ernakulam and Alappuzha which are also the districts with the largest percentage of wetlands compared with their land area.

PUD counts were found to be significantly correlated (*Pearson's* $r = 0.84$, p-value < 0.01) with visitation in Kerala for 2014 based on the calibration model for 42 sites. The results of the OLS log-log regression, shows that PUD counts are able to explain 64% of the variation in the dependent variable. The hypothesis of normal distributions of the residuals is not rejected by the Shapiro-Wilk test. The results of the naïve, optimal, and smearing estimates of visitation based on the univariate calibration model are presented in Table 2. Of the three estimates, the Duan smearing estimate was associated with the lowest RMSE and MAE, and was thus implemented for the inference of visitation estimates. This resulted in an estimate of 13.8 million recreational visits to the wetlands of Kerala in 2014 (Fig. 3).

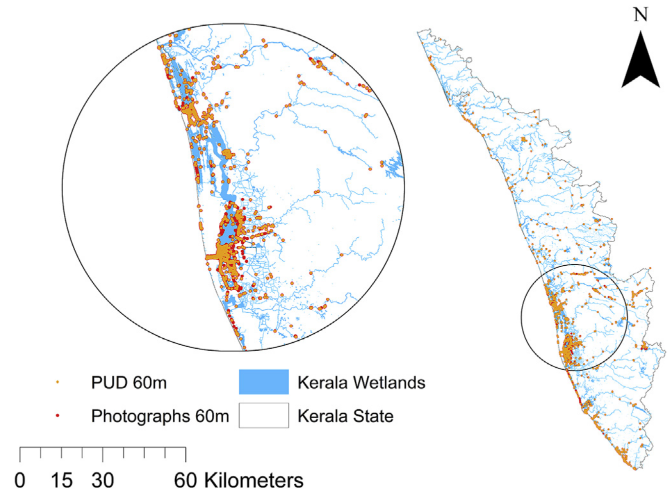

Fig. 2. Flickr photographs and PUD within 60 m buffer from Kerala wetlands. Callout shows the Vembanad Ramsar wetland area.

Table 2
Mean squared error of naïve, optimal, and smearing estimates from calibration model of wetland visits (n = 42).

Verification statistic	Naïve	Optimal	Smearing
RMSE	232,284	321,243	228,093
MAE	143,399	172,236	143,304

3.2. Water quality in Kerala's wetlands

Trends in water quality in Kerala are presented in Fig. 4 based on data compiled from the KSPCB. Water quality is deteriorating between 2008 and 2016 for all wetlands. The mean water quality in inland open waters has deteriorated from 2.1 to 2.8, in rivers from 2.4 to 4.1, and in coastal lagoons from 3.0 to 5.8 during 2008 and 2016. Fig. 5 presents estimated visitation density (visits per km^2) for each water quality class which appears to take a multimodal distribution. Visitation decreases once water quality deteriorates past the threshold for swimming (WQ = 2), then increases to its peak at class five, which is suitable for irrigation and controlled waste disposal, and decreases once water quality reaches the poorest standard of six which has no designated use assigned by the KSPCB.

3.3. Using geotagged photographs to model wetland quality

The results of the NB regression for wetlands in Kerala are shown in Table 3. Only statistically significant variables are retained (p-value < 0.05). The wetland area and wetland abundance variables show a positive correlation with PUD count, indicating that large sites surrounded by other wetlands attract on average higher visitation rates. The linear and quadratic terms of water quality have significant coefficients with opposite sign, indicating a concave functional form. Travel time to the nearest city has the expected negative relationship with PUD counts, indicating that visits reduce if a wetland is less accessible. Elevation has a positive relationship suggesting higher visitation rates in sites at high elevation.

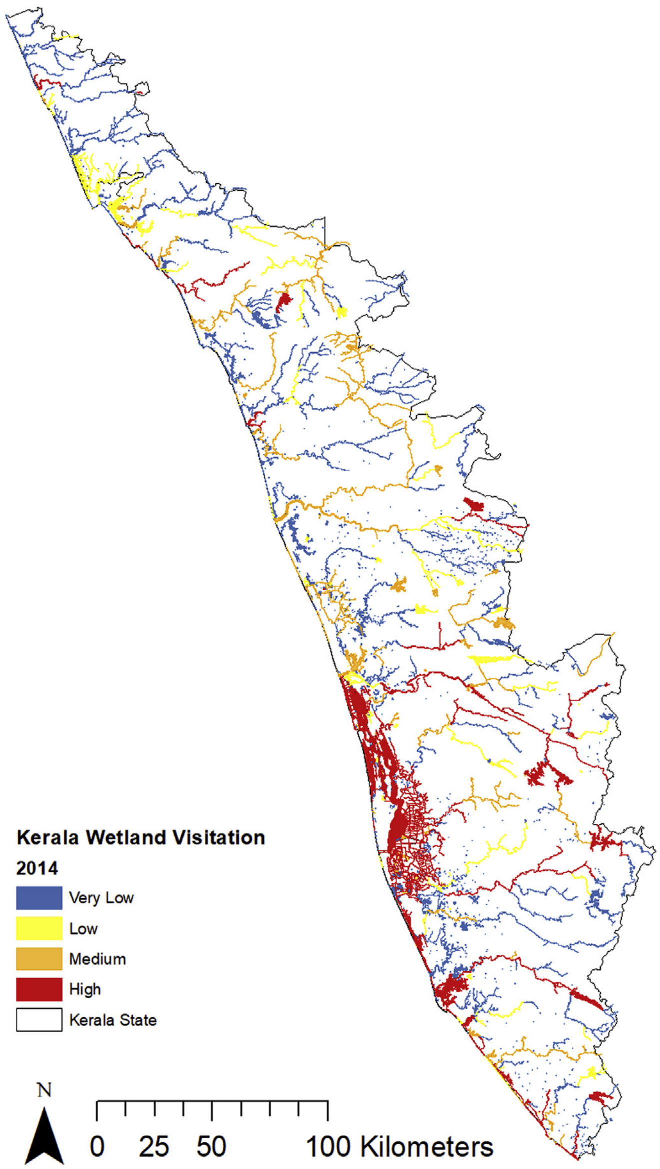

Fig. 3. Estimated intensity of recreation to wetlands in Kerala for year 2014 based on geotagged photographs from Flickr.

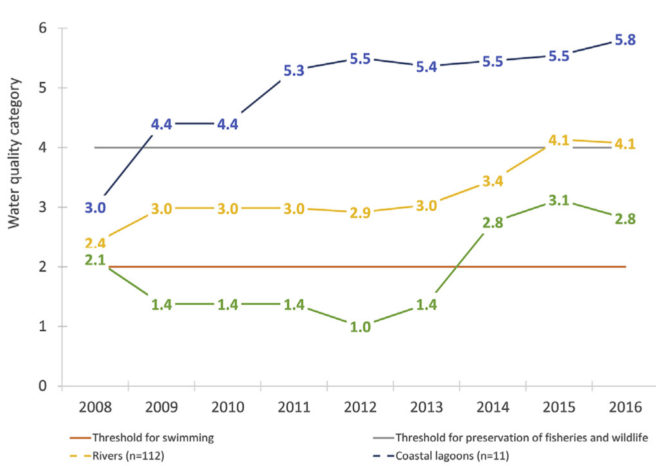

Fig. 4. Mean water quality in Kerala's wetlands (2008–2016) relative to limiting standards set by the KSPCB.

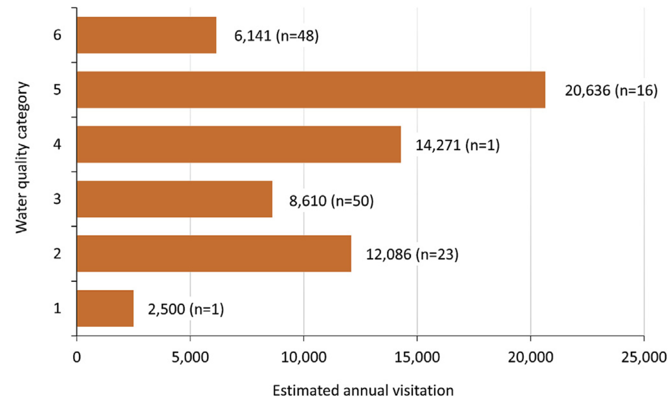

Fig. 5. Estimated average recreational visits for 2014 by wetland water quality class, as determined by the mode value between 2008 and 2016.

Table 3
Results of negative binomial regression of estimated visitation rates in 139 wetlands.

Parameter	Coefficient	Standard error
Intercept	−1.352	0.840
Wetland abundance	0.031***	0.008
AREA	0.040**	0.017
WQ	1.331***	0.517
WQ2	−0.159**	0.074
Travel time	−0.003**	0.002
Elevation	0.002**	0.001
(Scale)	1.053[a]	
(Negative binomial)	2.731	0.423
Log Likelihood	−344.4	
Adjusted Log Likelihood[b]	−327.2	
Akaike's Information Criterion (AIC)	704.8	

Notes: dependent variable = PUD60m; ** and *** denote significance at the 5%, and 1% level, respectively; [a] computed based on the deviance; [b] based on an estimated scale parameter and used in the model fitting omnibus test.

3.4. Assessing changes in wetland quality on recreation to Kerala's wetlands

Model estimates predict that improving the water quality to a minimum of class 4 in all sites would improve the tourist experience and result in a potential increase of 13% in the yearly visitation, amounting to almost 350,000 annual visitors. Model results also show this improvement is consistently positive across all selected wetlands, with visitor numbers increasing between 3% and 31% per year. Benefits are expected to be more substantial for wetlands presently classified as having the poorest water quality. The two Ramsar sites on the list, Vembanad Lake and Ashtamudi Lake, which are in the poorest water quality category, could, for instance, benefit by an additional 53,500 and 30,000 visitors per year. Decreasing wetland area appears to be more detrimental to visitation than increasing wetland area is beneficial to it. An increase or decrease in areal extent of 1% in all wetlands is estimated to results in a loss or gain of 0.6% in visitation. Restoration of 5% and 10% of wetland area is estimated to increase visitation by 3% and 6% respectfully, while further encroachment of 5% or 10% would result in losses of 3% and 7% in yearly visitation. The Vembanad and Ashtamudi Ramsar sites feature high in the list indicating their sensitivity to encroachment. Vembanad Lake, which has lost almost 7% in surface area (Nair and Babu, 2015), could benefit by an additional 7% in visitation if this area was restored. Ashtamudi Lake, which has recently lost 10% of its area (Sitaram, 2014), would also benefit by 7% if this encroached area were reclaimed. Projected visitation changes from changes in the areal extent of individual coastal lagoons and inland open waters in Kerala are shown in Table S1 in the Supplementary Material.

4. Discussion and conclusion

Passively crowdsourced data, such as geotagged photographs from social media, represent a valuable source of location-based and time-specific data which can have useful applications in environmental management and protection. Such data can be used as a source of revealed preferences behaviour and, supplemented by sources of secondary data, contribute to develop models which allow for human behaviour to be assessed relative to the natural environment and the ES provided by it.

Based on the analysis of metadata from geotagged photographs this research estimates wetland visitation for the entire state of Kerala and extends, for the first time at this scale to a developing region, the emerging application of ES modelling using data from social media and environmental quality data. The model was used to bring light to the impact of improved ecosystem quality on tourism and recreation, which is a sector of growing importance for the region. The development of this technique may provide a useful tool for understanding the benefits of ecosystems and managing the impacts of human activities thereupon in the context of the growing concerns about environmental sustainability in Kerala (Paul et al., 2014; Varkey et al., 2016) and, by extension, in other developing regions where natural capital is at risk.

Consistently with previous research (Ghermandi, 2016; Keeler et al., 2015; Wood et al., 2013) PUD counts were found to be a fairly accurate measure of visitation in Kerala and for scenic areas in particular. Keeler et al. (2015) reported adjusted R^2 values of 0.65 and 0.7 for Iowa and Minnesota lakes, while Wood et al. (2013) reported a value of 0.39 from 836 recreational sites worldwide. The calibration model for sites in Kerala, with an adjusted R^2 equal to 0.63, lies towards the upper limit of this range, supporting the notion that geotagged photo counts can be a proxy for nature-based visitation also in developing regions, where data scarcity is still a main concern. For comparison, Keeler et al. (2015) found on average 13.5 Flickr photos per lake within a 30 m buffer of Iowa and Minnesota lakes. For the present study, only 3.7 photos per wetland could be retrieved from within the same buffer, in spite of the fact that four more years were included in the search. In this context, the combination of different source of geotagged data (e.g., from Panoramio, Twitter or Instagram) has the potential to improve the model's geographic coverage and predictive power (Ghermandi, 2016; Tenkanen et al., 2017). Further refinement of the calibration model to separately analyse correlations for different subtypes of recreational sites (e.g., sites with scenic views, cultural interest, wetland ecosystems or protected areas) (Wood et al., 2013) and at a higher spatial resolution, such as monthly or daily visitation (Tenkanen et al., 2017) is currently limited by data scarcity but may become feasible as visitation data for Kerala becomes richer. At the time being, the availability of social media data and observed visitation from on-site monitoring for calibration purposes appear to be major limitations for the applicability of more sophisticated methods.

In spite of such limitation, this research demonstrates the ability to use social media data to provide visitation estimates in a data-scarce region at a large geographic scale and in the context of a developing country. The 13.8 million recreational visitors estimated for the wetlands of Kerala provide a benchmark that can be of importance to a range of stakeholders in the public and private sectors, and serve future management and conservation efforts, policy decisions, and forthcoming studies focussed on the regions wetland ecosystems. Further, these estimates reinforce the understanding of the crucial role that Kerala's wealth of unique wetland ecosystems plays in attracting tourists and recreationists. Comparing our estimates to the official visitation figures for Kerala (Kerala Department of Tourism, 2014), this study suggests that around 70% of recreation in Kerala can be attributed to its wetlands, with the majority of visitation concentrated in and around the Vembanad Lake coastal area, whose unique aesthetic appeal has received international recognition. These results seem consistent with the fact that the region's tourism industry is primarily driven by the draw of Kerala's rich natural environment.

Yearly visitation in Kerala has more than doubled since 2005 and the state harboured 800,000 more visitors in 2016 than in 2015. This indicates that Kerala's tourism industry is in a phase of rapid growth. The deterioration of water quality shown by Kerala's wetlands over the past decade calls into question wetland management and, importantly for the tourism industry, the impact of deteriorating water quality on recreational visitation if it continues to grow. The majority of the highly visited wetlands are also presently experiencing the lowest category of water quality. Over 40% of visitation occurs in wetlands in the lowest two quality categories, primarily in the coastal zone, with only 10% of visitation concentrated in wetlands experiencing the highest two quality categories, primarily in the upper reaches. The growth experienced at state level is not necessarily consistent at individual wetland level. Data sourced from local councils around the Ashtamudi Lake Ramsar site, for example, highlight the potential impacts of water quality on recreation. Until 2014, hotel stays showed an increasing trend of recreation to the Ashtamudi Lake. However, overnight hotel stays in 2014–15 stagnated at 30,000 and by 2016–17 had declined by 13%–26,000 while registered houseboat trips, which is a key activity on the lake, decreased from 958 to 610 during the same period. Ashtamudi Lake has been suffering under the lowest water quality class since 2011 and water quality continues to deteriorate, though its classification cannot be further downgraded as no lower threshold is currently recognized in Kerala. Although detailed information regarding the breakdown of visitors between local recreationists and tourists, as well as tourism and recreation-related expenditures are generally not available for individual wetlands, the economic valuation of recreation in the Vembanad Lake and the surrounding wetlands proposed by Sinclair et al. (2018) suggests that benefits are not limited to tourists but are substantial for local residents as well.

In this study we find a quadratic relationship between water quality level and visitation. This suggests that visitors are relatively insensitive to moderate water quality deterioration, but that visitation is negatively impacted once water quality deteriorates further, which appears to be the case in the Ashtamudi Lake. This is consistent with the primary recreational uses of the wetlands being for boating, and landscape and aesthetic appreciation, which are likely less sensitive to water quality deterioration than other activities such as swimming. Insofar as visitation by recreation is concerned, the highest benefits from water quality improvement are thus to be expected in wetland sites experiencing the poorest water quality. Important focus areas for conservation efforts are the two Ramsar wetlands, Ashtamudi and Vembanad, which are both experiencing some of the poorest water quality in the region. Some positive signals for the future management of the two sites come from the Integrated Management Planning Frameworks (IMPFs) for their future preservation (WISA, 2013; WISA & CWRDM, 2017). The IMPFs for the conservation and wise use of Vembanad and Ashtamudi Lakes highlights wetland pollution and lake shrinkage as key issues for the sustainable management of wetlands in Kerala. In response to deteriorating wetland quality the IMPFs have detailed the necessary budgets for the restoration and sustainable management of each wetland. For the Vembanad project, 1.4 billion rupees are requested for water quality management and biodiversity conservation and 500 million rupees towards the development of ecotourism around the Ramsar site (WISA, 2013). Similarly, for the Ashtamudi Lake, 1.7 billion rupees are requested for the water management and biodiversity conservation within which ecotourism is a major focus (WISA & CWRDM, 2017).

The detrimental impact of water quality should not overshadow the negative impact of wetland areal loss and encroachment for Kerala's wetlands. The results of this research are the first to provide estimates of how attempts to preserve or restore lost wetland areal extent in Kerala could affect visitation. If action is taken to ensure that wetland area is restored, the tourism industry, as well as local stakeholders, should see positive economic returns from increased visitation. Though changes to wetland area are not feasible in the short to mid-term they can be formed as part of long term strategies which these finding should help to support (Sitaram, 2014). Future research should seek to understand the costs associated with such strategies, to perform the appropriate cost benefit analyses in support of the decision-making process. Comparative analyses of the costs of improved environmental management (such as those proposed in Vembanad and Ashtamudi Lakes' IMPFs) should take into account the full extent of multiple uses of the wetlands that would be impacted, which, in addition to tourism and recreation, encompasses support for fisheries, aquaculture, carbon sequestration and flood protection as well as passive values (Anoop, Suryaprakash, Umesh, & Amjath Babu, 2008; Anoop & Suryaprakash, 2008).

The results of the study are not without limitations. First, there is a need to understand better the demographic of Flickr users in the sample, which may be biased towards specific sectors of the population (e.g., young, male and educated, foreign rather than local visitors) as have been noted for social media users in general (Nielsen, 2006). Techniques may be developed to account for such biases, if present, but their development so far has been limited due to the restricted information about socio-demographic details of Flickr and other social media users (Heikinheimo et al., 2017). To add to the complexity, demographic profiles of visitors to natural areas are often missing, especially in developing regions where even basic visitation data are often lacking. Second, although the technique based on PUD counts was used in several previous studies (Ghermandi, 2016; Keeler et al., 2015; Wood et al., 2013), there may be an added value in further extending the analysis to include an automated assessment of the content of the images, using for instance tools which have recently become commercially available thus enabling to identify and characterize different types of recreational activities (Richards & Tuncer, 2018). It should be noted, however, that the information provided by such tools is still relatively limited and that their use would substantially increase the costs of the analysis while at the same time raise ethical questions about accessing the actual content of a user's photograph. Third, considering that Kerala lies within the monsoon region, water quality shows substantial seasonal variation. Including seasonal changes in the analysis would be useful to understand how they impact visitation. This was however not possible in the present research since seasonal water quality information is not uniformly available for all sites in Kerala. Moreover, the inclusion of additional sites-specific data regarding for instance the presence of tourism facilities in the visitation regression model would be beneficial for the analysis but could not be performed due to lack of information for the entire network of wetlands in the state of Kerala.

Despite its limitations and although in its infancy, passive crowdsourced data from social media has, in our opinion, the potential to revolutionise the way we analyse nature-based tourism and recreation and cultural ecosystem services in general. Models developed using crowdsourced data, supported by secondary data, have the potential to complement, or potentially replace, resource-dependant and scope-limited primary field-based research techniques, to provide results at broad spatial scales and at limited cost. A better understanding of human behaviour in the context of the natural environment is fundamental to support adequate conservation decisions at the relevant policy scales, (e.g., regional or national) which is of critical importance especially to developing nations, whose rich natural assets are often faced with rapidly evolving socio-economic drivers and pressures which, if not properly managed, may severely impact future welfare and prosperity of their inhabitants.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tourman.2018.10.018.

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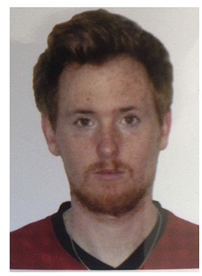

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